Project: Struct-X

CSD361(Introduction to Machine Learning)

Overview:

Struct-X is a computer-vision and machine learning-based project for estimating the structural vibration from an event camera without physical contact. The model is based on real recorded event-camera data and laser vibrometer data from the XJTU-DV-Beam dataset. The primary research thrust is robustness; that is, whether a trained model can estimate vibration under varying illumination or event-camera settings.

Problem Statement:

Structural vibration is an important signal for monitoring the condition of mechanical and civil structures. Traditional sensors such as accelerometers and laser vibrometers can provide accurate measurements, but they require physical placement or line-of-sight instrumentation. Event cameras provide asynchronous measurements of brightness changes and are well suited to high-speed motion and vibration because of their very high temporal resolution and low motion blur. However, event-based measurements can change substantially with sensing conditions such as illumination and camera parameters.

The project therefore asks a focused question: Can a deep spatiotemporal model learn the relationship between event-camera observations and true structural vibration, while remaining robust when the sensing conditions change?

Important scope decision: this project is about vibration estimation and sensing robustness, not structural-damage classification. The selected Beam dataset does not provide damage labels. This avoids making a claim that the dataset cannot be supported.

Base Paper + Links:

- **XJTU-DV dataset / companion paper:** X. Li, S. Yu, X. Chen, Y. Lei, B. Yang, N. Li, "XJTU-DV: Open-Source Dynamic Vision Dataset for Non-Contact Vibration Measurement and Fault Diagnosis of Mechanical Systems," Chinese Journal of Mechanical Engineering, 2026. Link
- **Dataset download:** XJTU-DV open dataset record (Zenodo), including XJTU-DV-Beam.zip. Link
- **Event vision:** G. Gallego et al., "Event-Based Vision: A Survey," IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022. Link
- **Spatiotemporal Transformer:** G. Bertasius, H. Wang, L. Torresani, "Is Space-Time Attention All You Need for Video Understanding?" ICML, 2021. Link
- **PINNs:** M. Raissi, P. Perdikaris, G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," Journal of Computational Physics, 2019. Link
- **Open-set SHM:** M. Zhou, Z. Lai, "Structural damage classification under varying environmental conditions and unknown classes via open set domain adaptation," Mechanical Systems and Signal Processing, 2024. Link
- **Physics-guided SHM:** "Structural damage identification by using physics-guided residual neural networks," Engineering Structures, 2024. Link
- **Computer vision + PINN:** "Accurate structural displacement reconstruction from acceleration and computer vision measurements using physics-informed neural networks," Mechanical Systems and Signal Processing, 2025. Link
- **Event-based deformation:** "Event-based Visual Deformation Measurement," 2026. Link

Paper used in our proposal	Publication venue	What to enter
XJTU-DV: Open-source dynamic vision dataset for non-contact vibration measurement and fault diagnosis of mechanical systems	*Chinese Journal of Mechanical Engineering*	**Q1**
Event-Based Vision: A Survey	*IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*	**Q1**
Physics-informed neural networks: A deep learning framework...	*Journal of Computational Physics*	**Q1**
Structural damage classification under varying environmental conditions and unknown classes via open set domain adaptation	*Mechanical Systems and Signal Processing*	**Q1**
Structural damage identification by using physics-guided residual neural networks	*Engineering Structures*	**Q1**
Accurate structural displacement reconstruction from acceleration and computer vision measurements using physics-informed neural networks	*Mechanical Systems and Signal Processing*	**Q1**
Is Space-Time Attention All You Need for Video Understanding? (TimeSformer)	**ICML**	**A***